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How the Largest Economies Fund Debt Service: Interest, Refinancing, and Revenue in the Top Five Economies

Kilambi Research

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Abstract. The five largest economies by GDP pay for debt service almost entirely by refinancing maturing principal and covering interest from a mix of revenue and new borrowing. The cross-country comparisons mix vintages, and the reader should know this before the first table: the latest observations are the United States 2024, Germany 2024, India 2022, China 2021, and Japan FY2025 (budget). Using central-government fiscal series for the United States, China, Germany, and India, Japan Ministry of Finance budget data for Japan, and market data through September 2026, this paper measures how much of each government's revenue goes to interest, how much of each year's net new borrowing just pays the interest bill, and how much of gross issuance is refinancing. The 2021–2024 rate-hiking cycle doubled the US federal interest burden, from 10.9 percent of revenue in 2020 to 20.3 percent in 2024 (3.6 percent of GDP); 45 percent of 2023 US net borrowing just paid interest. India carries the heaviest load in the sample at 34.0 percent of revenue in 2022, and 93 percent of its 2015–2022 borrowing need was interest. Germany is the lightest at 2.5 percent of revenue. China is small at 3.4 percent of revenue in 2021, and its central bank eased while others tightened. Japan's general-account interest bill, JPY 10.5 trillion in the FY2025 budget (9.2 percent of spending, about 1.7 percent of GDP), is projected by the Ministry of Finance to reach JPY 16.1 trillion by FY2028; roughly 83 percent of Japanese government bond issuance is refinancing. In the United States, 2024 gross federal issuance of USD 29.3 trillion against about USD 2 trillion of net new borrowing implies roughly 93 percent of gross issuance refinanced maturing principal. The interest-funding mix has shifted toward debt: the United States funded 42 percent of its interest bill from revenue in the 1990s and essentially none since 2002; India moved from 30 percent to zero over the same span. Germany is the exception, funding 90 percent of interest from revenue in the 2010s before the pandemic deficits returned the share to zero. The mix follows the primary balance, not the interest bill. World Bank series cover the central government only and have material gaps for China and Japan, documented in Section 7. This is a retrospective analysis; it makes no forecast.

Keywords: sovereign debt, debt service, interest burden, fiscal space, refinancing
JEL codes: E62, H63, H68

Production note. This paper was produced by Kilambi Research in September 2026. Computations were performed in Python on the archived WDI pulls and MOF documents listed in the data appendix; figures were rendered with Kilambi Research's institutional chart style. Data were retrieved in September 2026 and are subject to revision by the underlying sources.

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1 Introduction

How do the largest economies pay for their debt? The question has two parts. First, interest: what share of government revenue is consumed by interest payments, and what share of each year's new borrowing exists only to pay interest? Second, principal: when bonds mature, is the principal repaid from revenue or refinanced with new bonds? The answers determine fiscal space, which is the room a government has to fund everything else it does.

The question matters now because the global rate-hiking cycle of 2021–2024 repriced sovereign borrowing costs after a decade of near-zero rates. Interest bills that were flat for years are rising, and the rise lands unevenly: it depends on debt stocks, maturity structures, and how much of the budget was already committed to interest. This paper measures the funding of debt service for the five largest economies by nominal GDP, the United States, China, Germany, Japan, and India, over 2000–2026 where data permit.

This paper does three things. It assembles central-government fiscal series from the World Bank's World Development Indicators (WDI) [The World Bank, 2026], retrieved in September 2026, and computes a consistent set of debt-service metrics: interest as a share of revenue, interest as a share of GDP, the primary balance, and the share of net new borrowing absorbed by interest. It adds market context from 10-year government bond yields and debt-scale benchmarks. And it treats Japan separately using Ministry of Finance (MOF) budget documents [Japan Ministry of Finance, 2025a, 2026], because WDI carries no Japanese central-government interest series and Japan's debt dynamics are extreme enough to need their own accounting.

The preview of results: the US federal interest burden doubled between 2020 and 2024, from 10.9 to 20.3 percent of revenue, and 45 percent of 2023 net borrowing just paid interest. India has the heaviest burden in the sample: 34.0 percent of revenue in 2022, with 93 percent of its 2015–2022 borrowing need consisting of interest. Germany has the lightest: 2.5 percent of revenue. China's burden is small, 3.4 percent of revenue in 2021, and its central bank moved the opposite direction from Western central banks. Japan's interest bill is snapping back after three decades of decline: JPY 10.5 trillion in the FY2025 initial budget, JPY 13.0 trillion in the FY2026 budget, a MOF projection of JPY 16.1 trillion by FY2028 [Japan Ministry of Finance, 2025b], and a Fiscal System Council high-yield scenario of JPY 45.2 trillion by FY2035 [Fiscal System Council (Japan), 2025]. Across all five, maturing principal is overwhelmingly refinanced rather than repaid: in the US, about 93 percent of 2024 gross issuance was refinancing.

Section 2 places the paper against the fiscal-space literature. Section 3 describes how sovereign funding works. Section 4 documents the data. Section 5 gives the exact computations. Section 6 presents results. Section 7 states the limitations. Section 8 concludes. A data appendix lists every series and source. This paper is a retrospective analysis of the 2000–2026 window; it makes no forecast.

2 Related literature

The question of how governments fund debt service sits inside the debt-sustainability literature, which this paper does not try to extend. Bohn [1998] tests whether the US primary surplus responds to debt, the core fiscal-reaction condition; Barro [1979] derives optimal debt dynamics under tax smoothing. Ball et al. [1998] frame the deficit gamble: when growth exceeds the interest rate, debt

ratios can fall even with persistent deficits. Blanchard [2019] made that condition the center of the modern debate, arguing that low rates expand fiscal space. Ghosh et al. [2013] estimate fiscal space directly, modeling the primary-balance response to debt and the point of fiscal fatigue. Mauro et al. [2015] document two centuries of fiscal prudence and profligacy across countries, the long-run context for any single decade.

This paper's contribution is narrower and purely descriptive: it measures, for the five largest economies, the arithmetic of interest and refinancing over 2000–2026 from published fiscal series. It estimates no fiscal reaction function, computes no debt-sustainability threshold, and takes no position on optimal debt. Where the literature asks whether debt paths are sustainable, this paper asks what the current funding mix is. The interest-funding mix of Section 6.6 is closest in spirit to Ghosh et al. [2013]: both decompose how much of the fiscal effort goes to debt service, but this paper's split is accounting, not estimation.

3 Institutional background

Sovereign debt service has two components: interest on outstanding debt, and principal that matures and must be repaid or refinanced. In practice, large sovereign borrowers almost never repay principal from revenue. They refinance it: when a bond matures, they issue a new bond and use the proceeds to pay the old one. The standard measure is therefore gross issuance versus net issuance. Gross issuance is everything sold in a year, including bonds that replace maturing ones. Net issuance is gross minus maturing principal, which equals the year's new borrowing. When gross issuance is many times net issuance, the government is rolling its debt stock, and the interest bill is the part of debt service that actually consumes fiscal resources.

Interest, unlike principal, must be paid from revenue or from additional borrowing. Two ratios capture the burden. Interest as a share of revenue measures how much of the budget is precommitted before any policy choice is made. Interest as a share of GDP scales the burden to the economy. A third measure, the share of net new borrowing absorbed by interest, answers a narrower question: of the new debt issued in a year, how much exists only to pay interest? When that share approaches 100 percent, the government's entire deficit is the interest bill, and any primary deficit, spending beyond interest in excess of revenue, adds on top.

The primary balance, revenue minus non-interest spending, separates the interest burden from current policy choices. A government can run a primary surplus and still borrow, if interest exceeds that surplus. India has done exactly this for most of the sample period.

One definitional point matters throughout. The WDI fiscal series used here cover the *central* government, not general government [The World Bank, 2026]. For the United States that means the federal government, excluding states and municipalities. For Germany it means the Bund, excluding Länder. Debt and deficit figures from the Institute of International Finance (IIF) [Institute of International Finance, 2024], used for scale, are general-government concepts and are therefore larger. The two are not interchangeable, and this paper does not mix them in any single computation.

Japan requires separate treatment. The WDI interest series has no Japanese observation at all, and Japan's debt stock, about 224 percent of GDP in Q4 2024 on IIF data [Institute of International Finance, 2024], is far outside the range of the other four. The Japanese analysis uses the general-account budget, the part of the budget the Diet votes annually, and MOF bond-issuance plans, which distinguish new bonds from refunding bonds [Japan Ministry of Finance, 2025a].

4 Data

4.1 Sources

Table 4 lists the sources. The fiscal core is the World Bank WDI [The World Bank, 2026], retrieved via the ElkassabgiData API [ElkassabgiData, 2026] in September 2026 (the World Bank API is the underlying source; ElkassabgiData is the retrieval interface). Six indicators are used: interest payments as a percent of revenue (GC.XPN.INTP.RV.ZS), revenue excluding grants as a percent of GDP (GC.REV.XGRT.GD.ZS), expense as a percent of GDP (GC.XPN.TOTL.GD.ZS), central-government debt as a percent of GDP (GC.DOD.TOTL.GD.ZS), tax revenue as a percent of GDP (GC.TAX.TOTL.GD.ZS), and net lending/borrowing as a percent of GDP (GC.NLD.TOTL.GD.ZS, used only to validate the balance proxy). All are central-government concepts in current local currency units, converted to ratios.

Bond yields are annual averages of 10-year government bond series: the US Federal Reserve Board H.15 series, the German Bundesbank series, the Sveriges Riksbank series for Japan, and the IMF International Financial Statistics series for India, all retrieved through the ElkassabgiData library [ElkassabgiData, 2026]. No Chinese 10-year series was available in the library; the paper uses the People's Bank of China policy rate path instead.

Debt scale comes from the IIF Global Debt Monitor for Q4 2024 [Institute of International Finance, 2024]. US gross issuance comes from SIFMA's 2024 fixed-income statistics [Securities Industry and Financial Markets Association, 2024]. Japan fiscal data come from MOF budget documents: the FY2025 budget fact sheet [Japan Ministry of Finance, 2025a], the FY2026 initial budget [Japan Ministry of Finance, 2026], the January 2025 MOF estimate reported by Reuters [Japan Ministry of Finance, 2025b], and the Fiscal System Council's high-yield scenario [Fiscal System Council (Japan), 2025]. Historical Japanese actuals are from the Board of Audit of Japan [Board of Audit of Japan, 2025].

Evidence grades. Every input is graded on two axes: derivation (Observed: measured directly; Reconstructed: arithmetic on stored series; Modelled: derived under assumptions; Assumed: judgment) and source quality (A: official statistics; B: reputable secondary compilations; C: unverifiable or web research). The WDI fiscal series are Observed/A; IIF debt Observed/B (general-government concept, Q4 2024); SIFMA issuance Observed/B; MOF budgets and Board of Audit actuals Observed/A; the MOF FY2028 projection and the Fiscal System Council FY2035 scenario are Assumed/B (official scenarios, not outturns); the balance proxy and the interest-funding mix are Reconstructed/A (documented arithmetic on Observed/A inputs).

4.2 Coverage and sample period

Table 1 shows coverage by indicator and country. The WDI panel is unbalanced. The United States and Germany have near-complete series from the early 1970s through 2024. India's interest and revenue series run 1974–2022 with a gap in 2019–2021. China's interest series covers only 2005–2021, and China has no WDI expense series at all, so no balance or primary-balance computation is possible for China. Japan has no WDI interest series and almost no revenue series; it is analyzed from MOF sources only. Expenditure, needed for the balance, ends in 2023 for the US and 2022 for Germany and India, so balance-based metrics end one to two years before the interest series. The interest-funding mix of Section 6.6 needs revenue, expense, and interest together; it covers the

United States for 1990–2023, Germany for 1990–2022, and India for 1990–2018 and 2022. The WDI central-government debt series is sparse outside the US and is not used in any computation; debt scale uses IIF throughout.

The analysis window shown in figures is 2000–2026. Computations use all available years. Data vintage: September 2026.

Table 1: Fiscal data coverage by country and indicator (WDI, central government)

Indicator	United States	China	Germany	India
Interest, % of revenue	1972–2024	2005–2021	1972–2024	1974–2022*
Revenue, % of GDP	1972–2024	2005–2024	1972–2024	1974–2022*
Expense, % of GDP	1972–2023	n/a	1972–2022	1974–2022*
Tax revenue, % of GDP	1972–2024	2005–2024	1972–2024	1974–2022*

Notes: *India has no WDI observations for 2019–2021. Japan is excluded: WDI carries no Japanese central-government interest series. *Source:* The World Bank [2026] via ElkassabgiData [2026], retrieved September 2026.

4.3 Summary statistics

Table 2 reports summary statistics for the core fiscal indicators over the 2000–2026 analysis window, computed from the archived September 2026 pull. The dispersion is the point: interest absorbs anywhere from 1.0 percent of revenue (Germany's minimum) to 40.9 percent (India's maximum). Japan has no usable WDI central-government revenue or interest series and is analyzed from MOF sources in Section 6.4.

Table 2: Summary statistics: central-government fiscal indicators, 2000–2026 window

Country	Indicator	Obs.	Mean	Min	Max
United States	Interest, % of revenue	25	11.8	8.8	20.3
United States	Revenue, % of GDP	25	18.0	15.7	20.6
United States	Expense, % of GDP	24	23.1	17.9	31.8
China	Interest, % of revenue	17	3.3	2.2	4.4
China	Revenue, % of GDP	20	13.5	9.5	16.5
Germany	Interest, % of revenue	25	4.0	1.0	6.9
Germany	Revenue, % of GDP	25	28.8	27.7	30.0
Germany	Expense, % of GDP	23	29.3	27.4	32.7
India	Interest, % of revenue	20	28.7	22.8	40.9
India	Revenue, % of GDP	20	12.2	9.4	14.4
India	Expense, % of GDP	20	15.6	13.7	17.2

Notes: Computed from the archived September 2026 WDI pull (`wdi_fiscal_long.csv`); years with no observation are skipped, none imputed. China's revenue maximum (16.5, 2007) and India's revenue minimum (9.4, 2022) are the series breaks flagged in Section 7. Japan: WDI carries no usable central-government revenue or interest series. *Source:* The World Bank [2026] via ElkassabgiData [2026]; author's computations.

Table 3 reports the latest available observation for each country and indicator, the basis for the

cross-country comparisons in Section 6. The vintages differ: United States 2024, Germany 2024, India 2022, China 2021, Japan FY2025 budget.

Table 3: Latest available debt-service indicators (mixed vintages)

Country	Year	Interest % of revenue	Interest % of GDP	Govt debt % of GDP†
United States	2024	20.3	3.6	120
China	2021	3.4	0.5	92
Germany	2024	2.5	0.7	60
Japan	FY2025	9.2‡	1.7	224
India	2022	34.0	3.2	83

Notes: †IIF Global Debt Monitor, Q4 2024, general-government concept; not comparable to the central-government ratios. ‡Percent of general-account spending (MOF FY2025 initial budget), not of revenue. Vintages differ by country by construction; see Section 4. *Source:* The World Bank [2026] via ElkassabgiData [2026]; Japan Ministry of Finance [2025a]; Institute of International Finance [2024].

5 Methodology

All computations are arithmetic on the series in Table 4; no statistical estimation is performed. For each country c and year t with interest share of revenue $i^{rev}_{c,t}$ and revenue share of GDP $r_{c,t}$:

$$i^{gdp}_{c,t} = \frac{i^{rev}_{c,t}}{100} \times r_{c,t} \tag{1}$$

which converts the interest burden to percent of GDP. The fiscal balance is proxied as revenue minus expense,

$$b_{c,t} = r_{c,t} - e_{c,t}, \tag{2}$$

where $e_{c,t}$ is expense as a percent of GDP. This is a net lending/borrowing proxy, not the WDI cash surplus/deficit series (GC.BAL.CASH.GD.ZS), which the World Bank has discontinued. The proxy is validated against the WDI net lending/borrowing series (GC.NLD.TOTL.GD.ZS): the two track within half a point of GDP in nearly every country-year, so the proxy is sound arithmetic, not an estimate. The primary balance adds back interest:

$$p_{c,t} = b_{c,t} + i^{gdp}_{c,t}. \tag{3}$$

Net new borrowing is $n_{c,t} = -b_{c,t}$ when $b_{c,t} < 0$, and zero otherwise. The share of net new borrowing absorbed by interest is

$$s_{c,t} = \frac{i^{gdp}_{c,t}}{n_{c,t}}, \tag{4}$$

defined only in deficit years. Readings above 100 percent mean the government ran a primary surplus and still had to borrow to cover interest. The cumulative borrowing mix sums interest paid and primary deficits from 2015 through the latest year with complete data and reports interest as a share of the two combined. The interest-funding mix splits the interest bill itself between revenue and new borrowing. Revenue pays non-interest spending first; whatever is left of a primary surplus, if any, pays interest. With $i^{gdp}_{c,t}$ and $p_{c,t}$ as defined above,

$$i^{rev}_{c,t} = \min(i^{gdp}_{c,t}, \max(0, p_{c,t})), \qquad i^{debt}_{c,t} = i^{gdp}_{c,t} - i^{rev}_{c,t}, \tag{5}$$

where i^{rev} is the revenue-funded portion of interest and i^{debt} the debt-funded portion. The revenue-funded share is $i^{rev}_{c,t}/i^{gdp}_{c,t}$, capped at 100 percent. In primary-deficit years the share is zero by construction: all interest is borrowed, and the primary deficit is borrowed on top.

Bond yields are annual averages of the underlying monthly or daily 10-year series. Japan's interest history combines Board of Audit actuals (FY1992, FY1999, FY2005) [Board of Audit of Japan, 2025], MOF initial budgets (FY2023, FY2025, FY2026) [Japan Ministry of Finance, 2025a, 2026], the MOF January 2025 projection for FY2028 [Japan Ministry of Finance, 2025b], and the Fiscal System Council high-yield scenario for FY2035 [Fiscal System Council (Japan), 2025]. The FY2026 budget figure assumes the MOF's assumed interest rate rising from 2.6 to 3.0 percent. The yen figures are general-account interest payments in trillions of yen.

Table 4: Data sources

Item	Source
Fiscal series	World Bank WDI [The World Bank, 2026], central government (interest, revenue, expense), via ElkassabgiData API [ElkassabgiData, 2026], retrieved September 2026
10-year yields (US, Germany, Japan, India)	ElkassabgiData library [ElkassabgiData, 2026]: Fed Board, Bundesbank, Riksbank, IMF IFS; annual averages computed by author
China rates	PBoC policy rate path (no 10-year series in library)
Government debt, % of GDP	IIF Global Debt Monitor, Q4 2024 [Institute of International Finance, 2024]
US gross issuance	SIFMA, 2024 fixed-income statistics [Securities Industry and Financial Markets Association, 2024]
Japan budgets and projections	MOF FY2025 budget fact sheet [Japan Ministry of Finance, 2025a]; MOF FY2026 initial budget [Japan Ministry of Finance, 2026]; MOF January 2025 estimate, reported by Reuters [Japan Ministry of Finance, 2025b]; Fiscal System Council high-yield scenario [Fiscal System Council (Japan), 2025]; Board of Audit of Japan, historical actuals [Board of Audit of Japan, 2025]
US FY2024 deficit	Congressional Budget Office [Congressional Budget Office, 2024]

Notes: Evidence grades (Section 4): WDI Observed/A; IIF and SIFMA Observed/B; MOF budgets and Board of Audit actuals Observed/A; MOF FY2028 projection and Fiscal System Council scenario Assumed/B. *Source:* As listed.

6 Results

6.1 The interest burden

Table 5 and Figure 1 give the headline result. The US federal interest burden doubled in four years, from 10.9 percent of revenue in 2020 to 20.3 percent in 2024, or 3.6 percent of GDP. The increase is entirely a rate effect working through refinancing: the debt stock was already large, and as low-coupon bonds matured they were replaced at higher yields. Figure 2 pairs the burden with the cause. The US 10-year annual average rose from 0.89 percent in 2020 to 4.21 percent in 2024; Germany's went from negative (-0.24 percent) to 2.65 percent; Japan's from 0.00 to 0.89 percent and then 1.53 percent in 2025.

Table 5: Interest burden, latest year versus 2020

Country	Latest year	Interest % of revenue	Interest % of GDP	2020 interest % of revenue
United States	2024	20.3	3.6	10.9
China	2021	3.4	0.5	3.9
Germany	2024	2.5	0.7	1.0
India	2022	34.0	3.2	n/a*

Notes: *WDI has no Indian observation for 2020. Japan is in Section 6.4. Vintages differ by country (Section 4). *Source:* The World Bank [2026] via ElkassabgiData [2026]; author's computations.

India is the extreme case. At 34.0 percent of revenue in 2022, more than a third of central-government revenue is pre-committed to interest before a single rupee of discretionary spending. The level has been above 20 percent for the entire 2000–2022 window shown in Figure 1, peaking above 40 percent in the early 2000s. Germany is the opposite: the burden fell steadily as rates declined, bottoming near 1.0 percent of revenue in 2020, and has only begun to turn up, reaching 2.5 percent in 2024. China's burden is small and flat, 3.4 percent of revenue in 2021, and its monetary stance diverged: the PBoC eased while the Fed, the ECB, and eventually the Bank of Japan tightened.

6.2 How much new borrowing just pays interest

Table 6 reports, for the latest year with complete revenue and expense data, the decomposition of net new borrowing. In the United States in 2023, interest was 3.15 percent of GDP and net borrowing 6.94 percent of GDP, so 45 percent of every new dollar borrowed just paid the interest bill; the rest covered a primary deficit of 3.79 percent of GDP. In India in 2022, 73 percent of net borrowing was interest, and the primary deficit was only 1.13 percent of GDP: nearly the whole deficit was the interest bill. Germany in 2022 was at 16 percent.

Figure 3 plots the series from 2000. Three features stand out. First, readings above 100 percent are common for India and occurred for Germany in the 2000s: in those years the government ran a primary surplus and still borrowed, because interest exceeded the surplus. The debt was growing while the non-interest budget was in surplus. Second, the US series collapsed in 2020, not because interest fell but because the COVID deficit was enormous: interest was a small share of a very large borrowing number. By 2023 the share was back to 45 percent. Third, Germany's long decline

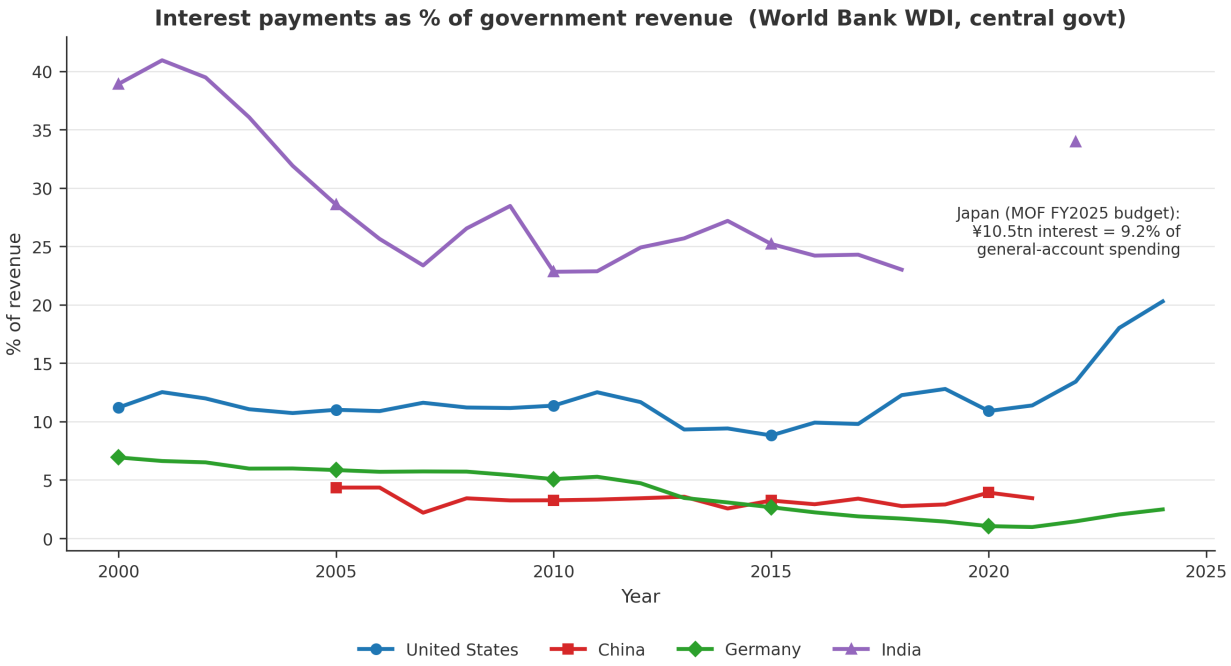


Figure 1: Interest payments as percent of central-government revenue, 2000–2024. India (purple) is persistently the highest, ending at 34.0 percent in 2022. The United States (blue) is flat near 10 percent until 2020, then rises steeply to 20.3 percent in 2024. Germany (green) falls to about 1 percent and ticks up to 2.5 percent. China (red) stays between 2 and 4 percent; its series ends in 2021. The Japan annotation is from the MOF FY2025 budget, not WDI. Source: World Bank WDI via ElkassabgiData; MOF.

Table 6: Interest share of net new borrowing, latest year with full data

Country	Year	Interest % of GDP	Net borrowing % of GDP	Interest share of borrowing	Primary balance % of GDP
United States	2023	3.15	6.94	45%	-3.79
Germany	2022	0.42	2.61	16%	-2.19
India	2022	3.19	4.35	73%	-1.16

Notes: Deficit years only. China has no WDI expense series; Japan is in Section 6.4. *Source:* The World Bank [2026] via ElkassabgiData [2026]; author's computations.

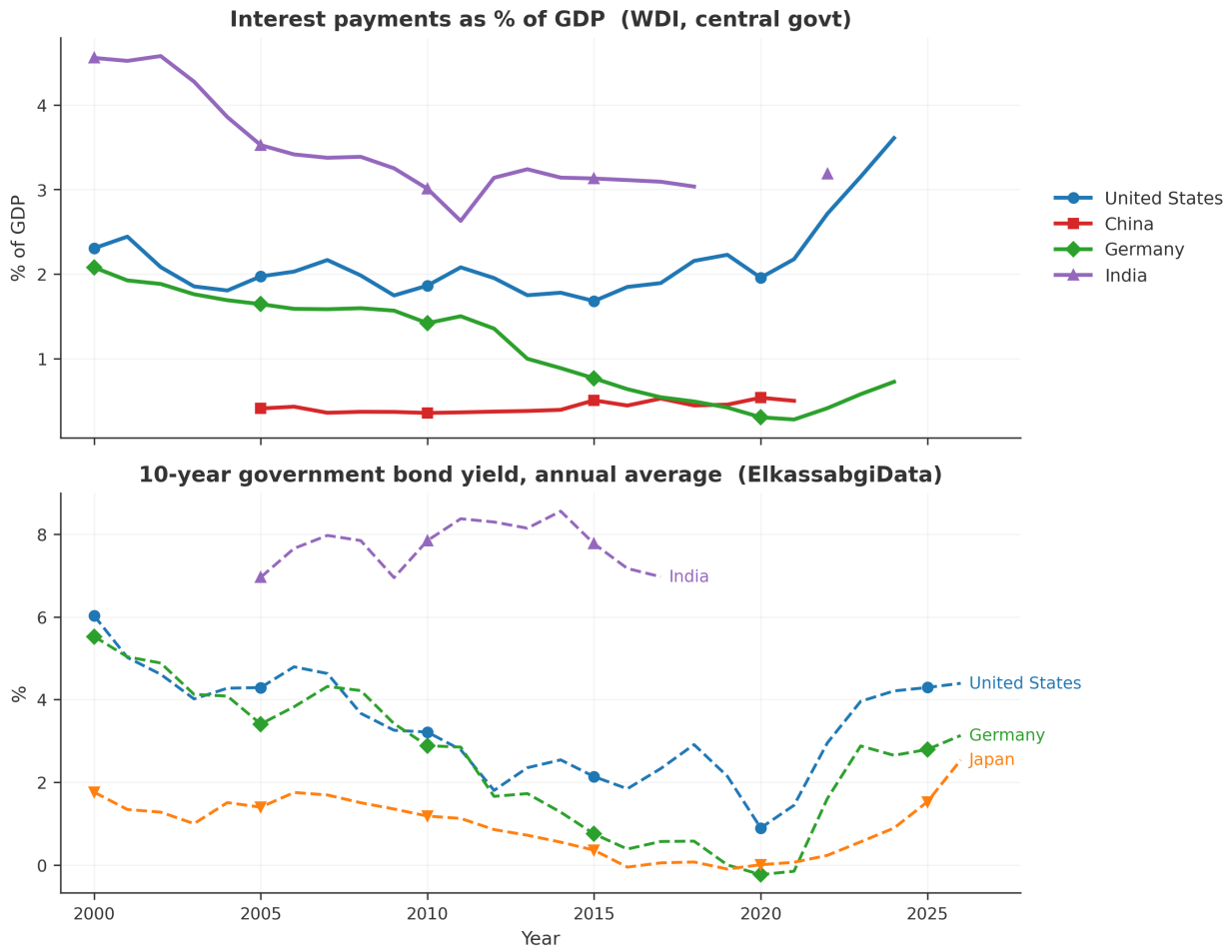


Figure 2: Top: interest payments as percent of GDP, central government. Bottom: 10-year government bond yields, annual averages. The US 10-year rose from 0.89 percent in 2020 to 4.21 percent in 2024; Germany from -0.24 to 2.65 percent; Japan from 0.00 to 0.89 percent. India's yield series ends in 2017 (7.17 percent in 2016). China has no 10-year series in the library; the PBoC cut its policy rate from 4.35 to 3.0 percent after 2016 while others hiked. Source: WDI via ElkassabgiData; ElkassabgiData yield library (Fed Board, Bundesbank, Riksbank, IMF IFS).

from above 100 percent to 16 percent is the fiscal dividend of the low-rate era, now reversing at the margin.

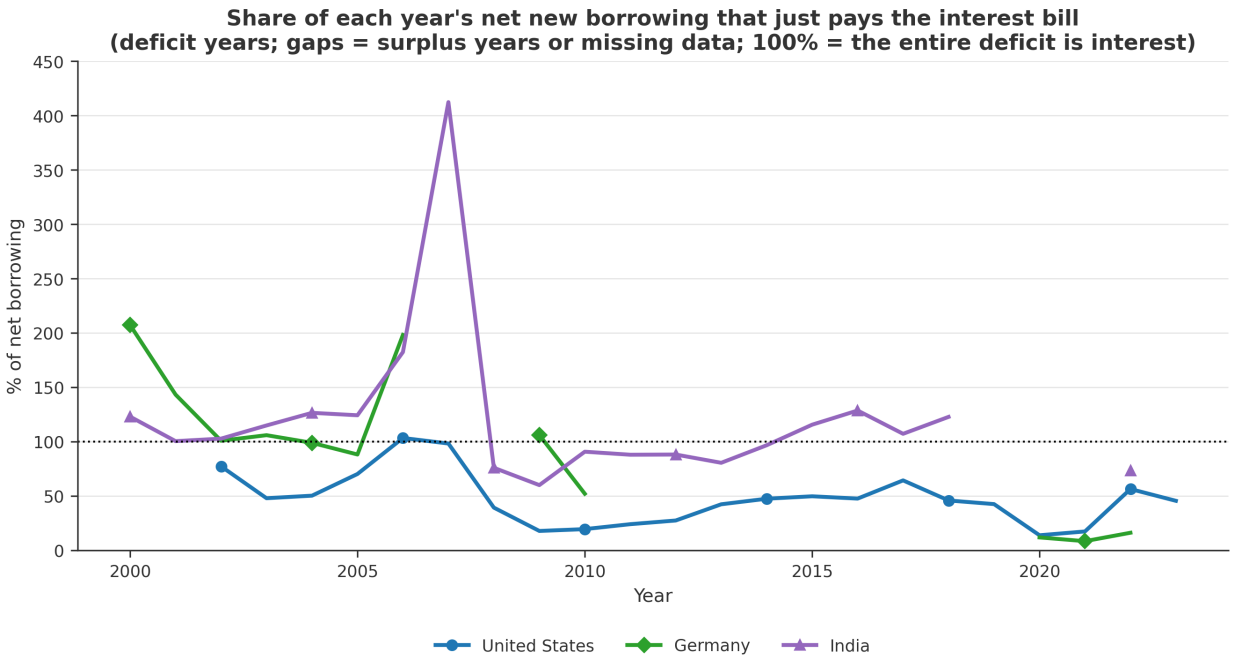


Figure 3: Share of each year's net new borrowing absorbed by interest payments, 2000–2023, deficit years only. The dotted line at 100 percent marks the point where the entire deficit is the interest bill; readings above it mean a primary surplus was insufficient to cover interest. India (purple) repeatedly exceeds 100 percent. The US (blue) share collapsed in 2020 on the COVID deficit, then recovered to 45 percent in 2023. Germany (green) declined from above 100 percent in the 2000s to 16 percent in 2022. Source: WDI via ElkassabgiData, author's computations.

6.3 The cumulative picture since 2015

Figure 4 (right panel) cumulates interest paid and primary deficits from 2015 through the latest complete year and reports interest as a share of total borrowing need. For the United States over 2015–2023, interest was 34 percent of all borrowing need (19.8 percent of GDP in cumulative interest against 39.2 percent of GDP in cumulative primary deficits). Germany over 2015–2022 is also 34 percent (3.9 against 7.7). India over 2015–2022 is 93 percent: cumulative interest of 15.6 percent of GDP against cumulative primary deficits of only 1.2 percent of GDP. India's debt growth since 2015 is, to a first approximation, the accumulation of its interest bill.

The left panel of Figure 4 gives the debt scale for context, from IIF Q4 2024 data [Institute of International Finance, 2024]: Japan 224 percent of GDP, the United States 120 percent, China 92 percent, India 83 percent, Germany 60 percent. These are general-government figures and are not directly comparable to the central-government ratios elsewhere in the paper; they are shown to fix the stocks behind the flows.

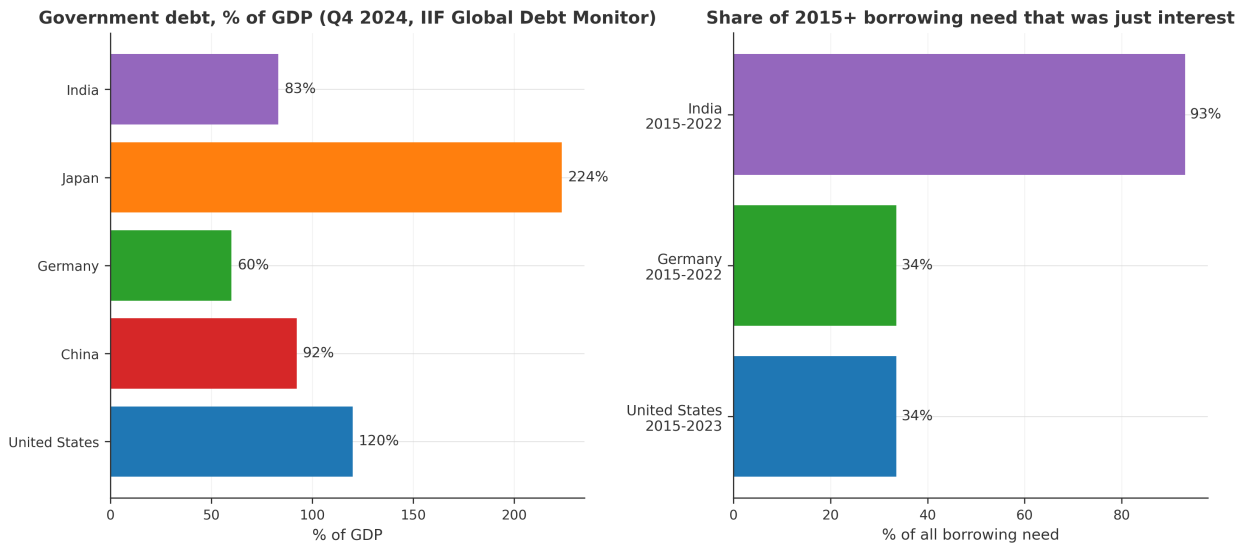


Figure 4: Left: government debt as percent of GDP, Q4 2024 (IIF, general-government concept). Right: interest as a share of total borrowing need, cumulated from 2015 through the latest year with complete data. India: 93 percent of 2015–2022 borrowing need was interest. The United States (2015–2023) and Germany (2015–2022): 34 percent each. Source: IIF Global Debt Monitor; WDI via ElkassabgiData, author's computations.

6.4 Japan: the long view

Japan's general-account interest payments trace the full arc of the low-rate era and its end (Figure 5). Actual payments were JPY 10.6 trillion in FY1992 and JPY 10.5 trillion in FY1999 [Board of Audit of Japan, 2025], then fell to JPY 7.0 trillion by FY2005 as rates declined, even as the debt stock grew. The FY2023 initial budget put interest at JPY 8.5 trillion; the FY2025 budget at JPY 10.5 trillion [Japan Ministry of Finance, 2025a], which is 9.2 percent of general-account spending and about 1.7 percent of GDP. The FY2026 initial budget reaches JPY 13.0 trillion on an assumed interest rate rising from 2.6 to 3.0 percent [Japan Ministry of Finance, 2026]. MOF's January 2025 estimate projects JPY 16.1 trillion by FY2028 [Japan Ministry of Finance, 2025b]. The Fiscal System Council's high-yield scenario, in which rates normalize further, reaches JPY 45.2 trillion by FY2035 [Fiscal System Council (Japan), 2025], more than four times the FY2025 level.

Two structural facts frame the Japanese numbers. First, roughly 83 percent of Japanese government bond issuance is refunding bonds, rolled principal, not new borrowing. The interest bill is the fiscal cost of the debt stock; the principal is perpetually refinanced. Second, the debt stock is about 224 percent of GDP (IIF, Q4 2024 [Institute of International Finance, 2024]). At that stock, each percentage point of the effective interest rate is worth more than 2 percent of GDP in annual debt service, which is why the MOF projections accelerate so fast once the assumed rate moves.

6.5 Refinancing: gross versus net issuance

The principal side of debt service is refinancing, everywhere in the sample. The cleanest quantification is the United States, where SIFMA reports 2024 gross federal issuance of USD 29.3 trillion [Securities Industry and Financial Markets Association, 2024] against a FY2024 federal deficit of

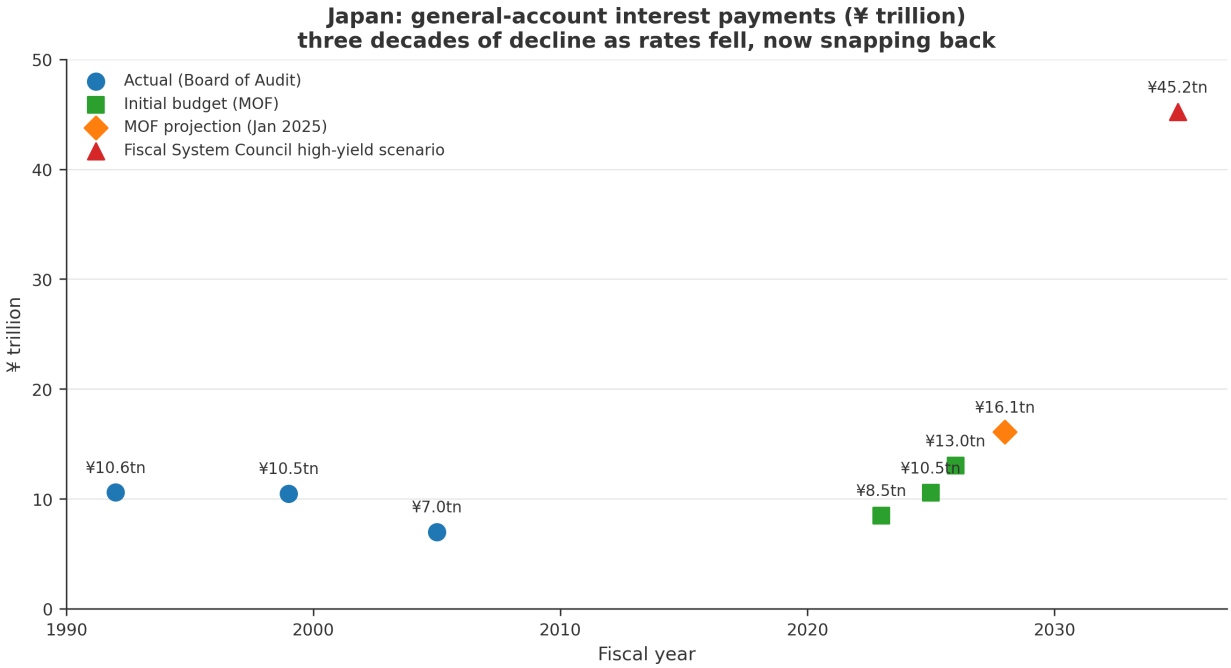


Figure 5: Japan general-account interest payments, trillions of yen. Actuals (Board of Audit): FY1992 JPY 10.6tn, FY1999 JPY 10.5tn, FY2005 JPY 7.0tn. Initial budgets (MOF): FY2023 JPY 8.5tn, FY2025 JPY 10.5tn, FY2026 JPY 13.0tn. MOF projection (January 2025): FY2028 JPY 16.1tn. Fiscal System Council high-yield scenario: FY2035 JPY 45.2tn. Three decades of decline as rates fell, now reversing. Source: Board of Audit of Japan; MOF; Reuters.

about USD 1.8 trillion (CBO [Congressional Budget Office, 2024]), the net new borrowing. Roughly 93 percent of gross issuance therefore refinanced maturing principal; about 7 percent was new debt. The same arithmetic holds qualitatively for the other four: no large sovereign repays principal from revenue at any scale. Debt service, as a fiscal burden, is the interest bill. The principal is a rollover operation, and its risk is refinancing risk, the risk that markets will not roll the stock at tolerable rates, not repayment risk.

6.6 The interest-funding mix: revenue versus new borrowing

The interest bill is paid from two sources: revenue left over after non-interest spending, and new borrowing. This section splits the bill between them, year by year, and asks whether the split has moved over time.

Figure 6 shows the annual split for the three countries where the decomposition is computable. Table 7 averages the revenue-funded share by decade. The answer is yes for the United States and India: interest used to be partly paid from revenue and is now paid entirely from borrowing.

The United States funded on average 42 percent of its interest bill from revenue in the 1990s, with full revenue funding in the 1997–2001 surplus years. The share fell to 20 percent in the 2000s and to zero in every year since 2002 except 2006, when a small primary surplus covered 3 percent of the bill. India's path is the same shape at a heavier burden: 30 percent revenue-funded in the 1990s, 20 percent in the 2000s, 7 percent in the 2010s, and zero in the latest observation. India never fully funded interest from revenue in the sample window.

Germany is the counterexample. Its revenue-funded share averaged 17 percent in the 1990s, 34 percent in the 2000s, and 90 percent in the 2010s, with the interest bill fully revenue-funded in 2007–2008 and every year from 2011 through 2019. The pandemic deficits snapped the share back to zero in 2020–2022. The German story is the same mechanism in reverse: the mix follows the primary balance, not the interest bill.

Table 7: Revenue-funded share of the interest bill, by decade (percent)

	1990s	2000s	2010s	2020–2024
United States	42	20	0	0
Germany	17	34	90	0
India	30	20	7	0

Notes: Decade means of the annual revenue-funded share, capped at 100 percent. Average interest bill (percent of GDP) and years covered: United States 3.05 (1990–1999), 2.04 (2000–2009), 1.92 (2010–2019), 2.50 (2020–2023); Germany 2.00, 1.73, 0.90, 0.34 (2020–2022); India 3.91, 3.87, 3.06, 3.19 (2022 only). China and Japan: decomposition not available. *Source:* Author's calculations from The World Bank [2026] via ElkassabgiData [2026].

The central point is that the shift toward debt funding did not happen because interest bills grew. US interest averaged 3.05 percent of GDP in the 1990s and 1.92 percent in the 2010s; India's averaged 3.91 and 3.06 percent. The bills shrank while the funding moved to debt, because primary balances deteriorated: revenue stopped covering non-interest spending, so nothing was left for interest. The mix is a primary-balance story, not an interest-rate story. Rates determine the size of the bill; the primary balance determines who pays it.

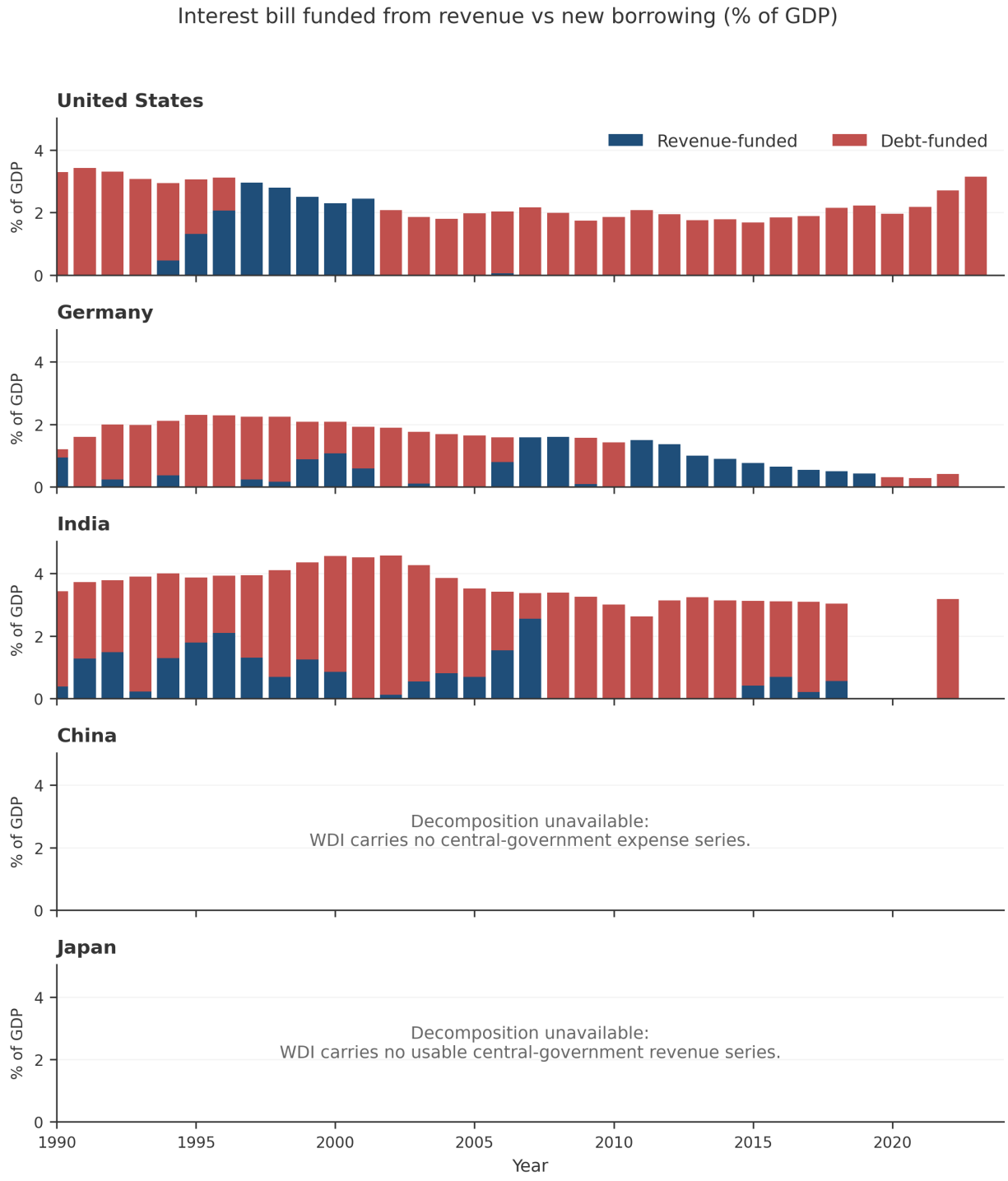
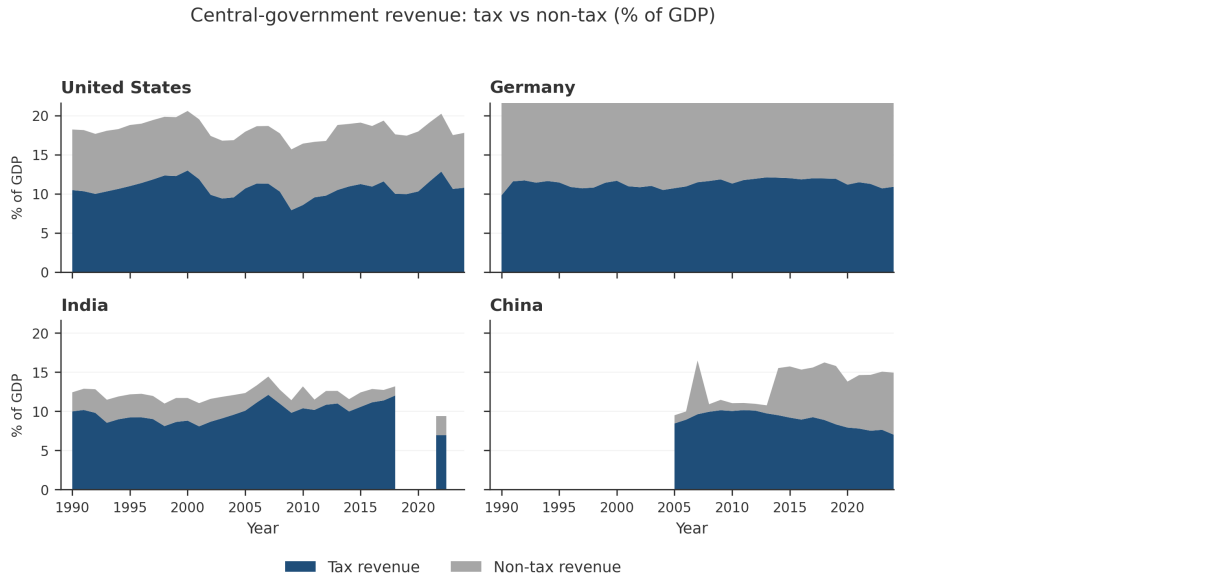


Figure 6: Annual interest bill split into revenue-funded and debt-funded portions, percent of GDP. China: WDI carries no central-government expense series, so no primary balance exists. Japan: WDI carries no usable central-government revenue series. Source: author's calculations from World Bank WDI via ElkassabgiData, retrieved September 2026.

6.7 Where the revenue comes from

Figure 7 splits central-government revenue into tax and non-tax components as percent of GDP. Three facts stand out. First, the level differs enormously across the sample. Germany's central government collects about 28–30 percent of GDP, roughly three-fifths of it non-tax, overwhelmingly social contributions (the WDI central-government concept includes social security funds). The United States collects 17–21 percent of GDP, about three-fifths of it tax. India collects 12–13 percent, about four-fifths tax. China collects 10–16 percent, with tax revenue steady near 8–10 percent of GDP.



Source: World Bank WDI (central government) via ElkassabgiData, retrieved September 2026. Non-tax = revenue excluding grants minus tax revenue (includes social contributions, property income, other).

Figure 7: Central-government revenue split into tax and non-tax components, percent of GDP. Non-tax is revenue excluding grants minus tax revenue and includes social contributions, property income, and other receipts. Japan: WDI revenue series has only three observations and is not shown. Source: World Bank WDI via ElkassabgiData, retrieved September 2026.

Second, the tax share is stable within each country over the full window. The US tax take moves with the cycle around 10–13 percent of GDP; Germany's sits near 11 percent; India's near 9–12 percent until the 2022 observation. What moves the total is the non-tax component, mainly social contributions in the United States and Germany.

Third, the broad categories behind the tax numbers are institutional background, not estimates from this paper. In the United States, individual income taxes are roughly half of federal receipts and social insurance contributions roughly a third, with corporate income tax under a tenth. In Germany, value-added tax and wage and income taxes are the two pillars, shared between the federation, the states, and municipalities. In India, goods and services taxes (since the 2017 GST reform), corporation tax, and personal income tax dominate Union revenue; customs and excise led historically. In China, value-added tax is the largest single tax and corporate income tax the second; individual income tax is a small share. The WDI tax-type detail series were checked and rejected: the US income-tax sub-series reports implausible levels and China's goes negative in 2005, so revenue composition is reported as tax versus non-tax only.

Two series breaks qualify the picture. China's 2007 revenue observation jumps to 16.5 percent of GDP from 10.0 percent the year before and returns to 10.9 percent the year after, with no matching move in tax revenue; it reads as a series break, not an economic event. India's 2022 observation, the first after the 2019–2021 gap, shows revenue at 9.4 percent of GDP against 13.2 percent in 2018 and tax revenue at 6.9 against 12.0 percent; the 34.0 percent interest share of revenue reported for 2022 in Section 6.1 rests on this observation and should be read with that break in mind.

7 Limitations and robustness

The WDI fiscal series cover the central government only. State, prefectural, and municipal debt and interest are excluded, so the burdens measured here understate general-government burdens, materially for Japan and Germany, where subnational borrowing is significant.

Coverage gaps are material. China has no WDI expense series, so no balance, primary balance, or interest-share-of-borrowing computation exists for China; the Chinese results are interest ratios only, ending in 2021. Japan has no WDI interest series at all and is analyzed from MOF budget documents, which are general-account concepts, not central-government WDI concepts; the two are not strictly comparable. India's WDI series skip 2019–2021, so no 2020 comparison is available for India. India's 10-year yield series ends in 2017, so the yield panel cannot show India's recent rate path. The interest-funding mix of Section 6.6 is computable only for the United States, Germany, and India; China and Japan are excluded for the reasons above.

The balance is proxied as revenue minus expense. The WDI cash surplus/deficit series (GC.BAL.CASH.GD.ZS) has been discontinued by the World Bank; the proxy is validated against the WDI net lending/borrowing series instead, which it tracks within half a point of GDP in nearly every country-year. For central governments the two track closely, but they can differ in years with large net acquisition of financial assets.

Two revenue series breaks deserve explicit flags. China's 2007 revenue observation (16.5 percent of GDP) is far above the surrounding years with no matching tax move and reads as a series break. India's 2022 observation, the first after the 2019–2021 gap, shows revenue at 9.4 percent of GDP against 13.2 percent in 2018; the 34.0 percent interest share of revenue for 2022 reported in Section 6.1 depends on that observation. The WDI tax-type detail series (income, goods and services, and other taxes as percent of revenue) were checked and not used: the US income-tax sub-series reports implausible levels and China's goes negative in 2005. Revenue composition is therefore tax versus non-tax only.

The Japanese FY2026 budget figure and the FY2028 projection come via MOF documents and Reuters reporting of MOF estimates; they are official projections, not outturns, and the FY2028 number predates the FY2026 budget. The Fiscal System Council FY2035 figure is a high-yield scenario, an illustration of rate sensitivity, not a forecast. The 83 percent refunding share is from MOF issuance plans and moves year to year.

All WDI data are as retrieved in September 2026 and are subject to revision by the World Bank and by national sources. Vintage note: the September 2026 ElkassabgiData redistribution carried a stale 2022 observation for India (revenue 9.1, tax revenue 6.7, expense 13.3 percent of GDP). The direct WDI pull from the same month and the current WDI series both report 9.4, 6.9, and 13.7; the paper uses these values throughout, and the corrected rows are flagged in the archived data files. The IIF debt figures are Q4 2024; the SIFMA issuance figure is full-year 2024.

7.1 What would overturn these findings

The headline ratios are arithmetic on published WDI and MOF series; they would be overturned if the World Bank or the Ministry of Finance revises the underlying fiscal series, which both do routinely. The interest-funding mix result, that the mix follows the primary balance rather than the interest bill, would fail if a recomputation on general-government series (rather than central-government) reversed the decade pattern; the paper's mechanism predicts it would not, because primary balances move the same way in both concepts. The Japanese FY2028 and FY2035 figures are official projections and scenarios, not findings: actual outturns will supersede them. The 93-percent refinancing figure for the United States is 2024-specific; a year with unusually light maturities would show a lower share without changing the mechanism.

8 Conclusion

The five largest economies fund debt service the same way: they refinance principal and pay interest from revenue supplemented by new borrowing. What differs is how much of the revenue interest consumes. The 2021–2024 hiking cycle doubled the US burden to 20.3 percent of revenue; India was already at 34.0 percent and has been above 20 percent for two decades; Germany, at 2.5 percent, shows what the low-rate era bought and what its end is starting to take back; China is small and easing against the global tide; Japan's bill, after thirty years of decline, is projected to rise by half again by FY2028.

The interest-funding mix has moved with the primary balance, not the interest bill: the United States and India now borrow every unit of interest, while Germany funded nine-tenths of its interest from revenue through the 2010s before the pandemic reversed it. The binding constraint in every case is the interest share of revenue and the interest share of new borrowing, not the debt stock as a number. A debt stock of 224 percent of GDP at near-zero rates cost Japan less, relative to its budget, than India's far smaller stock costs India today. When rates move, the arithmetic in this paper is the mechanism.

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A Data appendix

A.1 Series definitions

A.2 Derived series

For each country-year with complete inputs: interest as percent of GDP = int_rev/100 × rev_gdp; balance = rev_gdp − exp_gdp; primary balance = balance + interest % of GDP; net borrowing = −balance in deficit years; interest share of borrowing = interest % of GDP / net borrowing. Cumulative borrowing mix: sums from 2015 through the latest year with complete revenue and expense data. Interest-funding mix: revenue-funded interest = $\min(\text{interest \% of GDP}, \max(0, \text{primary balance}))$; debt-funded interest = interest % of GDP minus the revenue-funded portion; revenue-funded share = the ratio, capped at 100 percent. Revenue composition: tax revenue % of GDP and non-tax revenue % of GDP = rev_gdp − tax_gdp.

Table 8: WDI series used

Label	WDI code	Definition
int_rev	GC.XPN.INTP.RV.ZS	Interest payments, % of revenue
rev_gdp	GC.REV.XGRT.GD.ZS	Revenue excluding grants, % of GDP
exp_gdp	GC.XPN.TOTL.GD.ZS	Expense, % of GDP
debt_gdp	GC.DOD.TOTL.GD.ZS	Central government debt, total, % of GDP
tax_gdp	GC.TAX.TOTL.GD.ZS	Tax revenue, % of GDP
netlend	GC.NLD.TOTL.GD.ZS	Net lending (+)/net borrowing (–), % of GDP

Notes: Retrieved via ElkassabgiData (worldbank series), September 2026. debt_gdp is documented for completeness; the paper uses IIF debt figures for scale and does not compute from debt_gdp. netlend is used only to validate the revenue-minus-expense balance proxy. *Source:* The World Bank [2026] via ElkassabgiData [2026].

A.3 Source files

The local source files are the `econ_*.csv` pulls (one per country-indicator, ElkassabgiData WDI downloads, September 2026), the `wdi_fiscal_long.csv` panel, the `mix_*.csv` pulls (extended indicators: tax revenue, net lending/borrowing, and tax-type detail, the last checked and not used) and `mix_decomposition.csv` (the computed revenue-funded versus debt-funded split), and the yield CSVs in the bond-market-trends data directory. Japan figures use the MOF, Board of Audit, and Reuters documents listed in References. Replication code: `analyze2.py` (figures 1–4 and all WDI-derived tables), `japan_fig.py` (figure 5), `mix_analysis.py` (figures 6–7 and the interest-funding mix), `fetch_wdi.py` (WDI retrieval).